

PANDA Component Description: *watchdog*

Overview

The watchdog component implements a simple watchdog timer. Its timeout period is configurable through generics. The watchdog can be configured to generate a low state on an external signal or an interrupt if a timeout occurs before the watchdog is reset.

Generics

Name	Type	Default	Description
clk_frequency	integer	10'000'000	Clock frequency in Hz
period_ms	integer	1'000	Timeout period in ms

External Ports

Name	Type	Width	Description
tout_n	std_logic	1	External timeout signal (low if timeout occurs)

Registers

Reset Watchdog

Name: RST
Address: Base + 0x00

Name	RST
R/W	W
Reset Value	-

RST is not a real register. A write operation on this pseudo register resets the watchdog timer, clears the interrupt and sets *tout_n* high. The written value does not have any effect on these operations. *RST* returns a random value on read operations.

Configuration Register

Name: CFG
Address: Base + 0x01

Name	-	-	-	-	-	INTON	EXTON
R/W	-	-	-	-	-	R/W	R/W
Reset Value	0	0	0	0	0	0	1

EXTON: '1' → A low state on *tout_n* is generated if a timeout occurs
 '0' → *tout_n* remains high if a timeout occurs

INTON: '1' → An interrupt is generated if a timeout occurs
 '0' → No interrupt is generated if a timeout occurs

Tipps

If *tout_n* is connected to the *res_n* port of the system as shown in the graph, the system will be reset if a timeout occurs. The watchdog component is designed for this usage, therefore it is guaranteed that *tout_n* stays low for at least one clock cycle if it is connectes as shown in the graph.

